

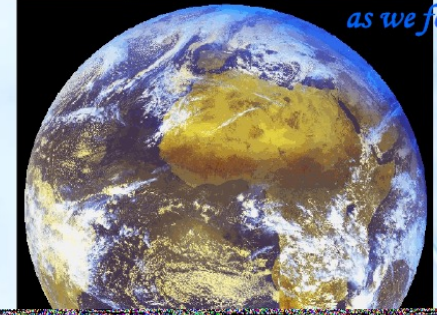
CHG 569

Environmental Pollution Engineering

Module V

(Strategies
for
Managing Air Pollution)

*Leave the earth for the next generation
as we found it....*



"The earth is our mother;
the earth does not belong to man, man belongs to the earth.
This we know - All things are connected like the blood which unites one family.
Whatever befalls the earth befalls the sons of the earth.
Man did not weave the web of life, he is merely a strand in it.
Whatever he does to the web, he does to himself."

"We do not inherit the earth from our ancestors;
we borrow it from our children"

Chief Seattle (1786 - 1866)
Patriarch of the Red Indians

What are the options available?



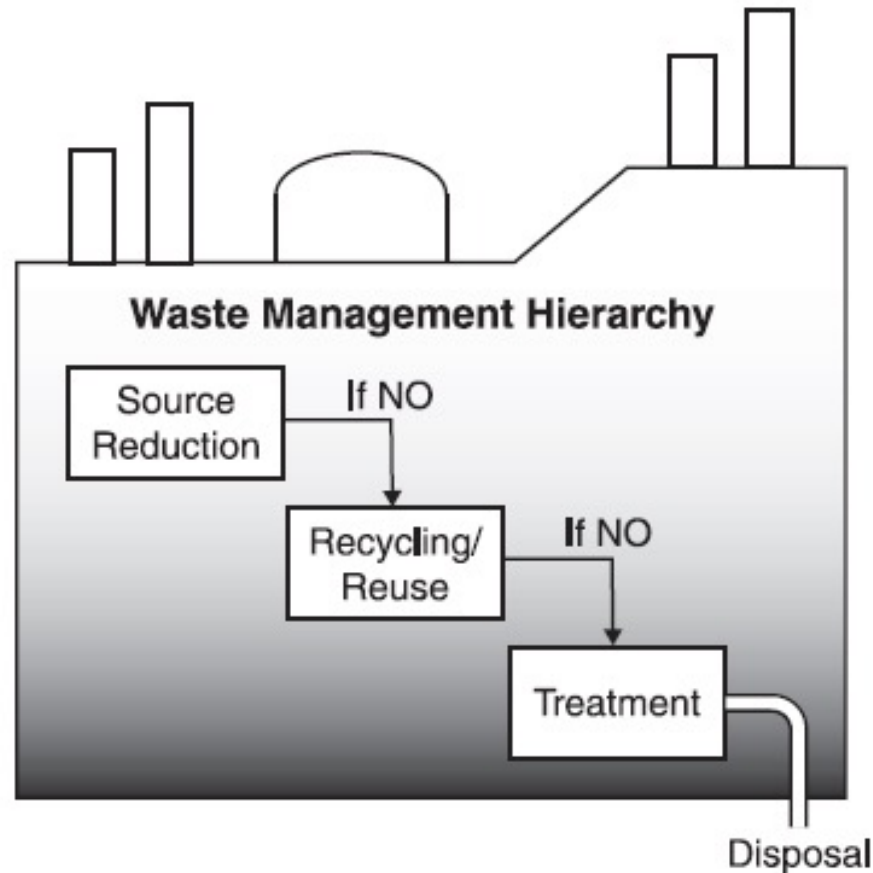
Source reduction – prevention & Reduction

Recycling – Reuse & Recycle

Benefit of the various options to the Environment



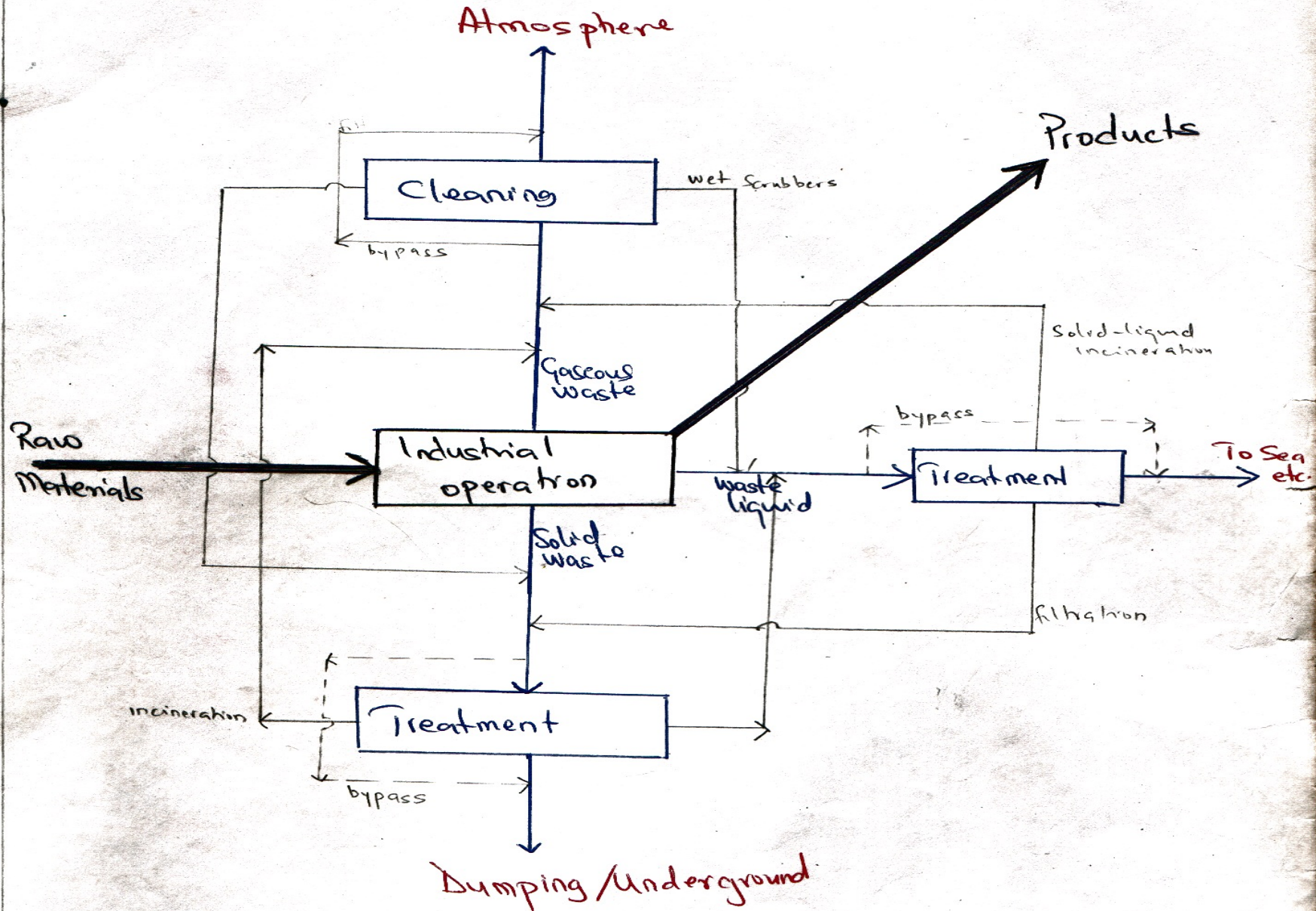
Waste Management Hierarchy



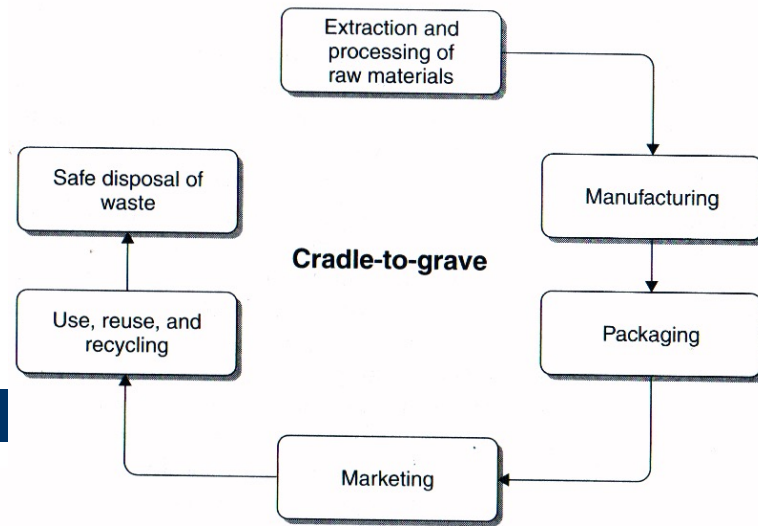
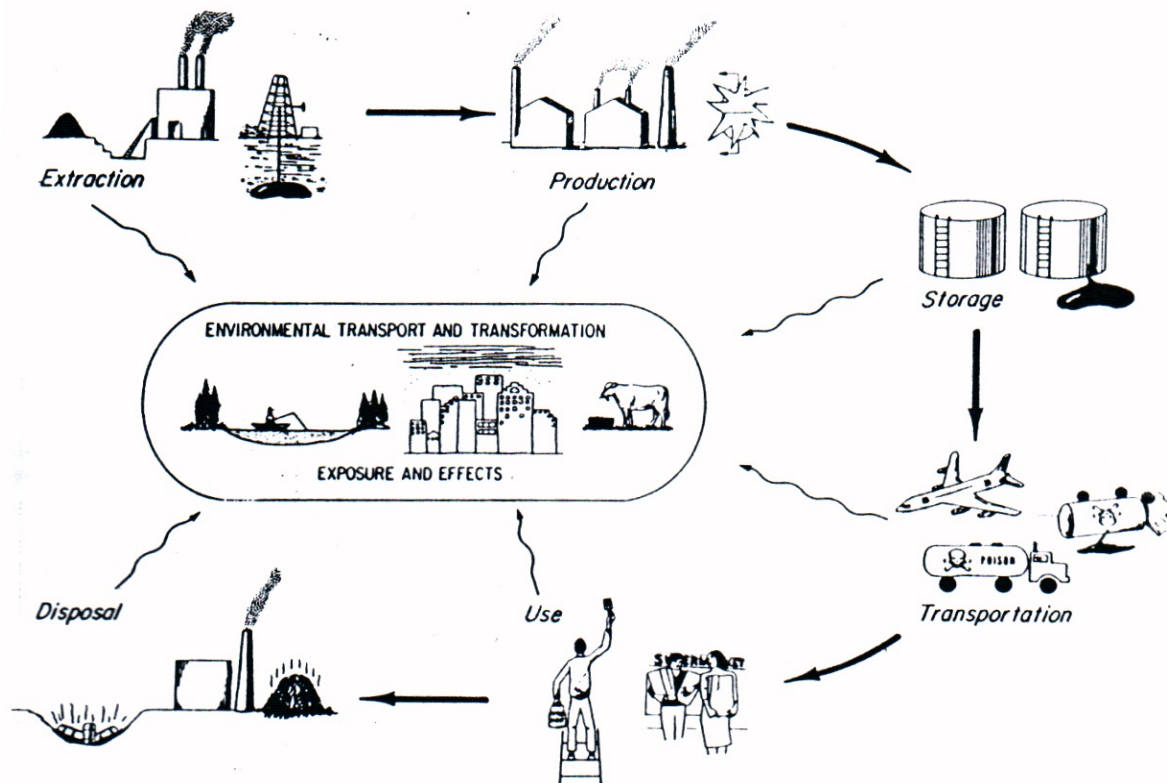
Approaches to solving the Environmental Problem

- **End-of-pipe approach (add-ons)**
 - Treatment at the end of the production process
 - After the event, react and treat approach
 - medicine after death approach
- **Cradle-to-grave approach**
 - Clean production
 - forward looking, anticipate and treat approach
 - Prevention is better than cure

The vicious cycle of end-of-pipe treatment



cradle-to-grave



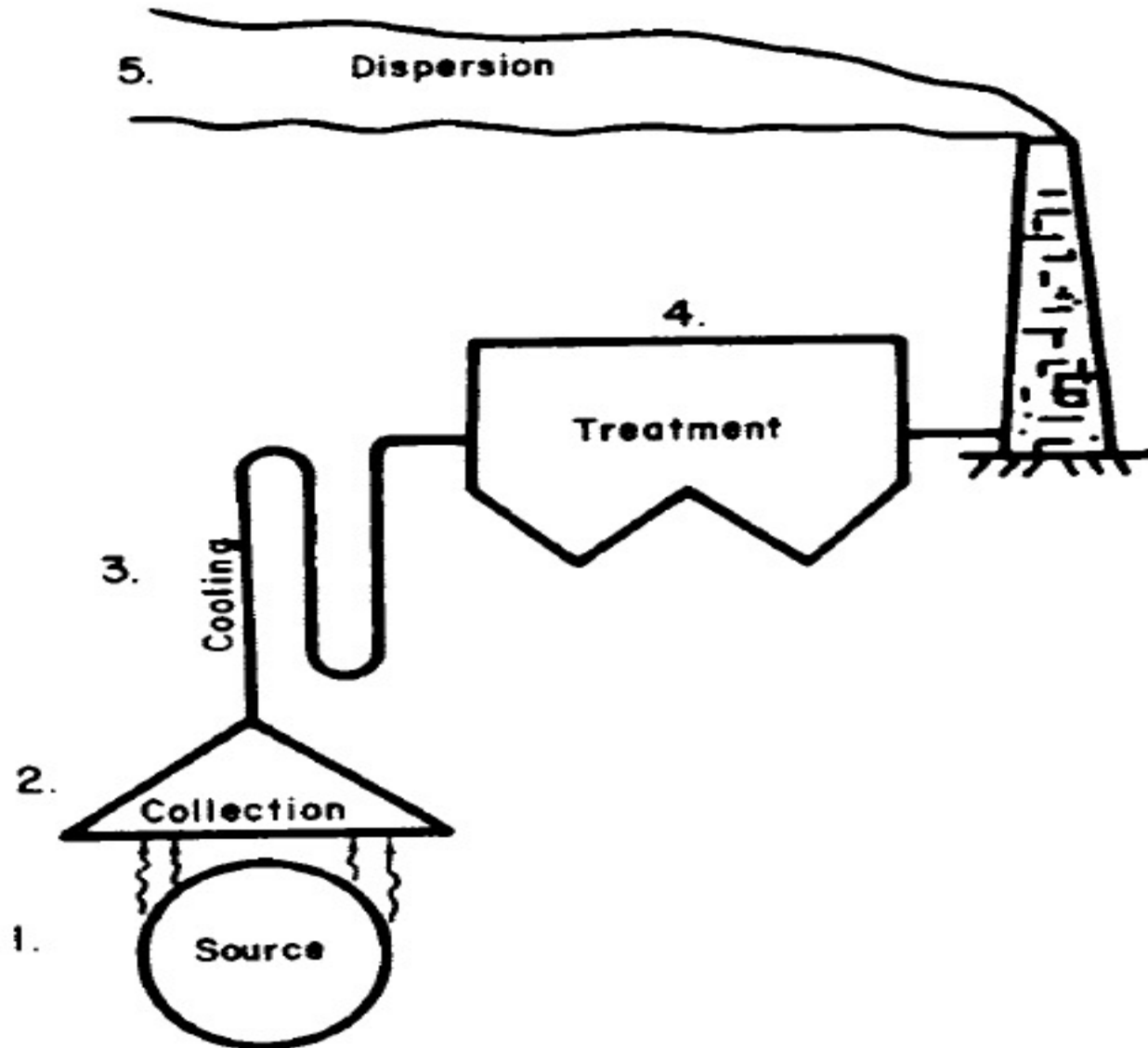
Differences in the two approach

End of pipe technology	Cradle-to-grave technology (Cleaner production)
how can we treat existing waste and emissions?	where do waste and emissions come from?
stands for re-action	stands for action
generally leads to additional costs	can help to reduce costs
waste and emissions are limited through treatment units storage of emissions	waste and emission prevention at the source avoids potentially toxic processes and materials
environmental protection comes in after products and processes have been developed	environmental protection comes in as an integral part of product design and process engineering
environmental problems are solved from a technological point of view	environmental problems are tackled at all levels / in all fields
environmental protection is a matter for competent experts	environmental protection is everybody's business
is bought from outside	is an innovation developed within the company
increases material and energy consumption	reduces material and energy consumption
environmental protection comes down to fulfilling legal prescriptions	environmental protection as a permanent challenge
is the result of a production paradigm dating from a time when environmental problems were not as yet known	is an approach intending to create production techniques for a more sustained development

Methods of Air Pollution Control

- Control at source – pollution prevention
- Abatement via treatment processes
- Dispersion (Dilution) of pollutant

Points of possible air pollution control



Methods of air pollution Control

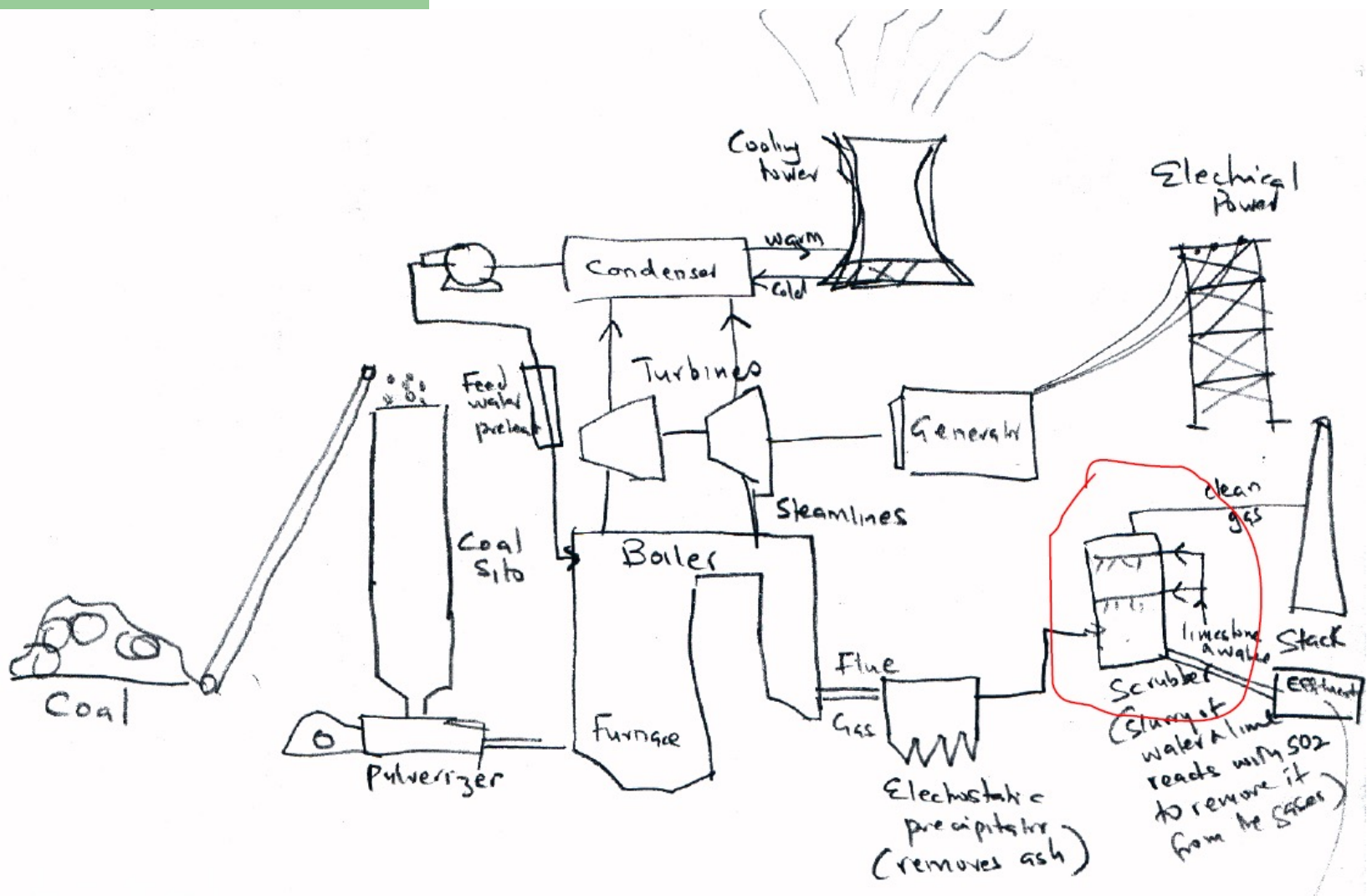
► Control at source (pollution prevention)

- Process change – Change in raw materials/ fuel /operating conditions/ or a completely new process.
 - conversion from a
 - power source using fossil fuel to one using solar or hydroelectric power -> less air pollution
 - high sulphur coal to low sulphur coal -> reduction in SO₂
 - replace coal with natural gas, a less polluting fuel.
 - Equipment modification
 - Replacing old and outdated plants with modernized facilities is also effective in reducing air pollution
- Good operating practices, such as good housekeeping and proper maintenance.
 - regular inspection and maintenance system to ensure that volatile organic compounds are not emitted by leaking equipment in a chemical plant.
 - can also result in cost savings to a plant by reducing losses of valuable materials.
- Plant shutdowns
 - This might be necessary in extreme cases, such as during an air pollution episode.

SO₂ Control thru Process Modification

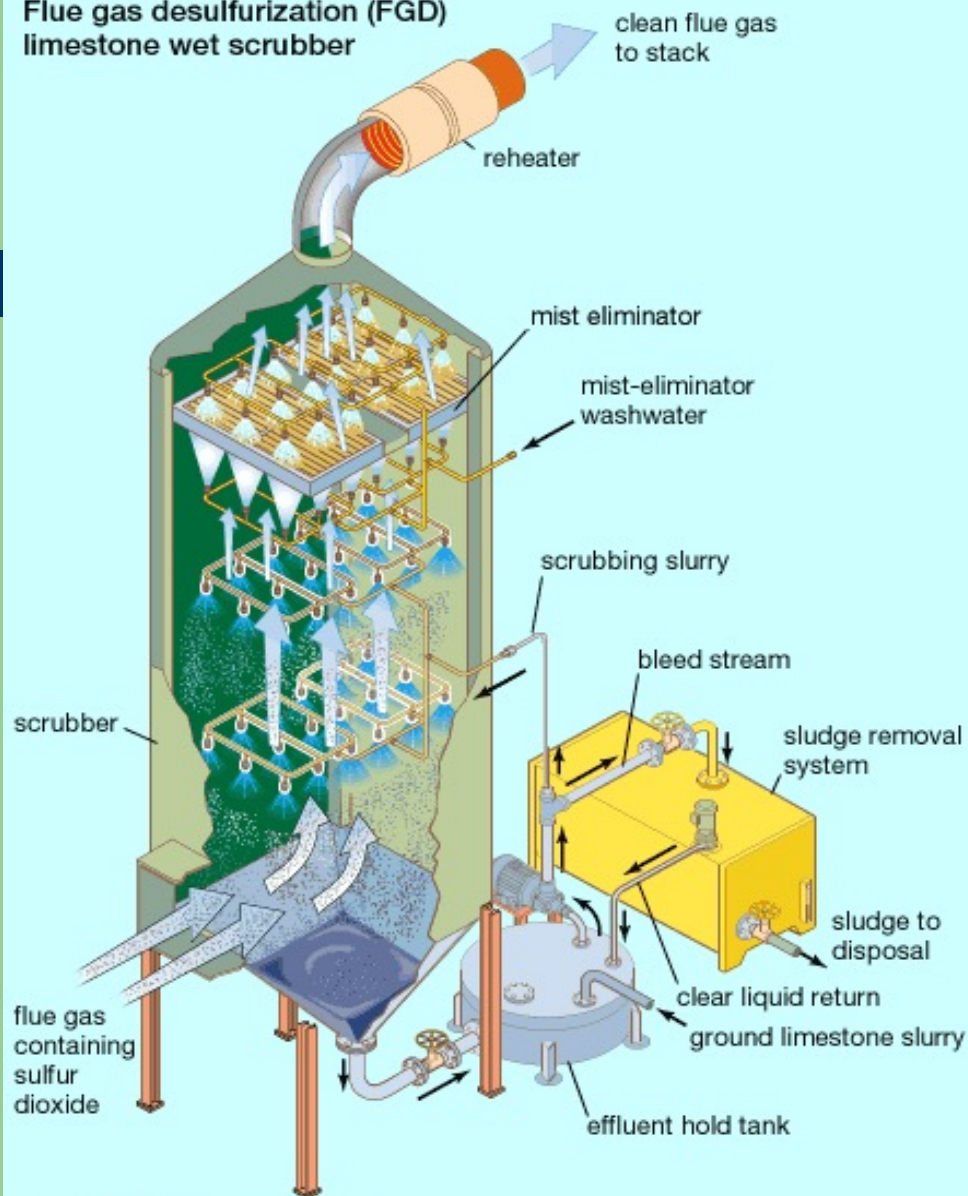
Flue gas desulphurisation (FGD)

used for removing SO₂ from the exhaust combustion flue gases of power plants that burn coal or oil to produce steam for the turbines that drive electricity generators.



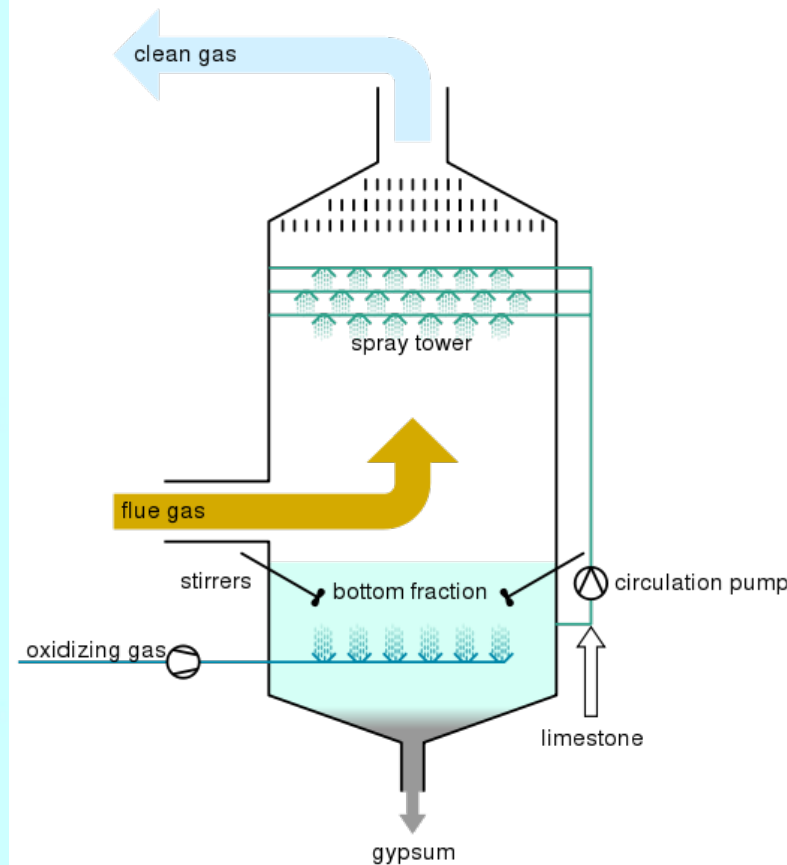
A coal-fired power plant using ESP for particulate and a limestone based SO₂ Scrubber

Flue gas desulfurization (FGD) limestone wet scrubber



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Absorber of the FGD



The reaction taking place in the absorber

Scrubbing using limestone (CaCO_3) slurry or lime (Ca(OH)_2) produces calcium sulphite- CaSO_3



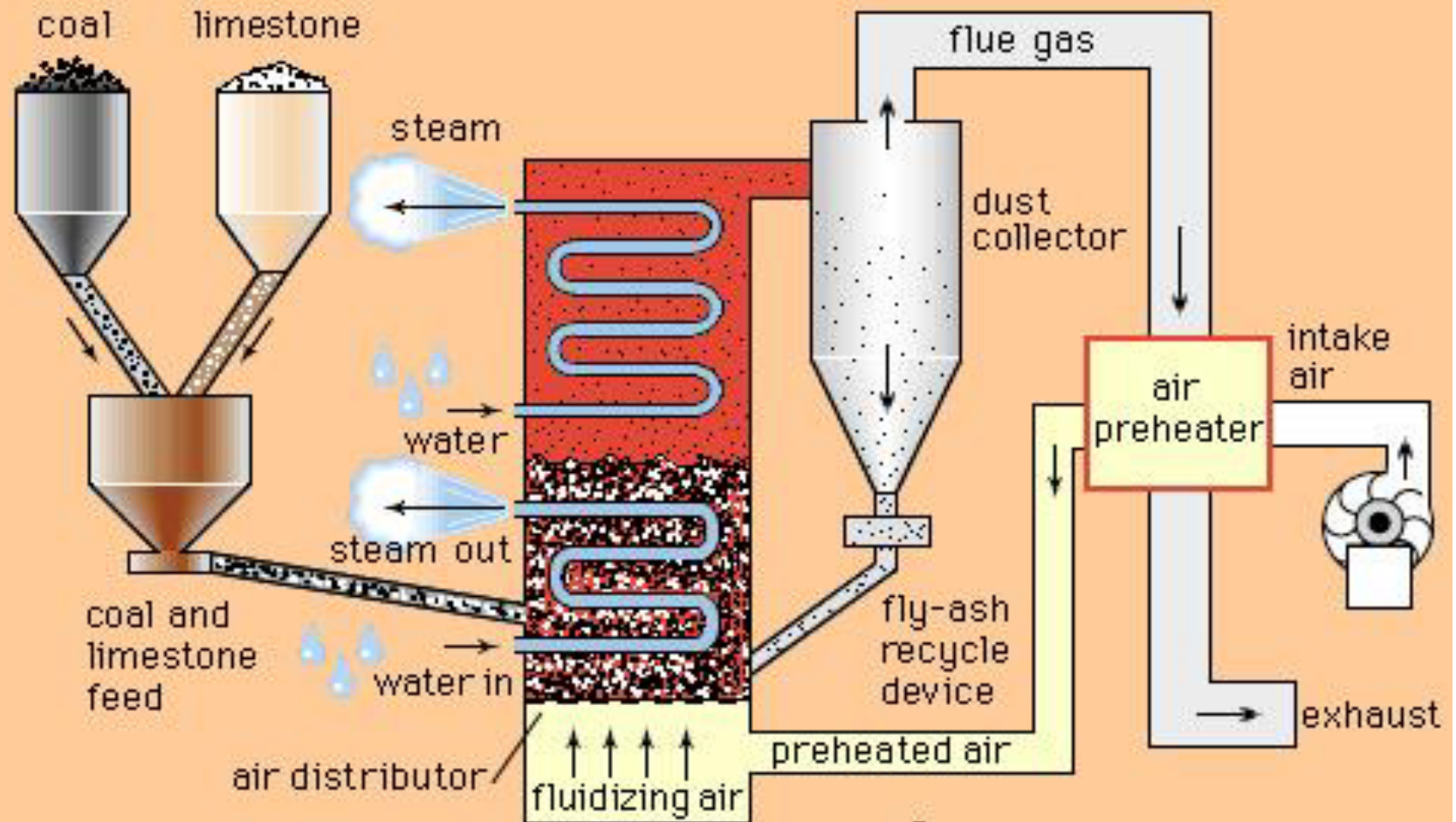
Some FGD systems go a step further and oxidize the calcium sulphite - CaSO_3 to produce marketable $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (gypsum):



Promising technologies

- Fluidized Bed Combustion
- Integrated Gasification Combined Cycle

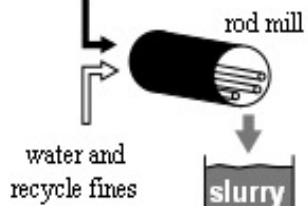
Fluidized Bed Combustion (FBC)



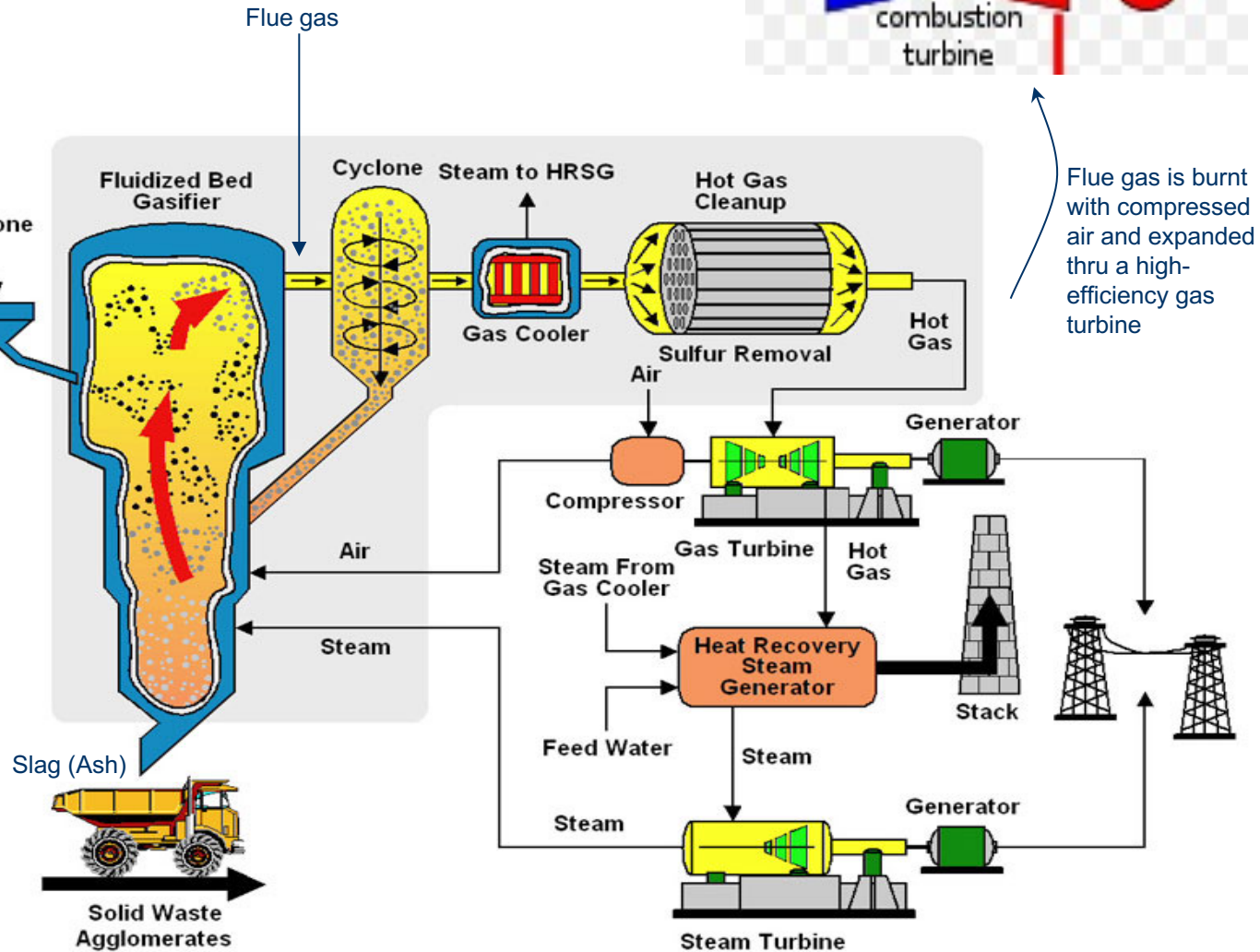
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IGCC

coal 2500 TPD



Coal & Limestone Handling



NO_x Control thru Combustion Modification

Various methods exist and these will be discussed as we go along, for now discussing control at source – pollution prevention methods

During combustion, NO_x formation is favoured by

- high peak temperature
- high residence time, and
- high oxygen concentration.

NO_x reduction can therefore be achieved through combustion modification techniques that

- Reduce peak temperature
- Reduce residence time
- Decrease oxygen concentration

To accomplish this, several techniques have been used:

Flue Gas Recirculation (FGR)

Cooled flue gas is recirculated and mixed with the combustion air entering the combustion chamber. This gas is deficient in oxygen and acts to reduce peak temperature (as would excess air) without increasing NO_x formation

with no recirculation, flame temperature = 3500°F

@ 20% FGR, flame temperature = 2900°F

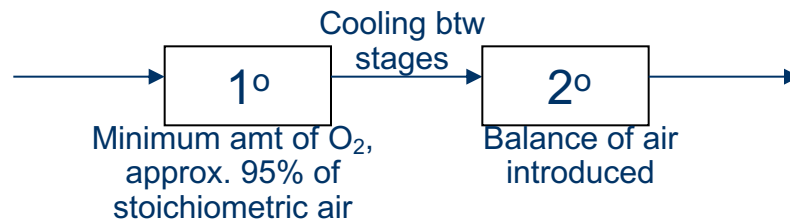
recirculation rates in the 15 – 20% range have been known to give NO_x reductions in the 40 – 55% range.

If flame temperature is held down to below 1,400°F, the thermal NO_x formation will be negligible

Staged Combustion (SC)

Staged or off stoichiometric combustion occurs in two stages (zones)

- Fuel in the 1st combustion area is fired with less than stoichiometric amount of air => fuel rich region
- Cooling between zones
- In the 2nd area the rest of the combustion air is introduced to complete fuel consumption.



**Deficiency of O₂
in the 1st zone**

+

**low temperature
in the 2nd zone**

=

**reduction in
NO_x formation**

Methods of air pollution Control

► Treatment (add-on controls)

- Gaseous pollutants (control devices to recover or destroy pollutant)
four commonly used control technologies for gaseous pollutants:
 - absorption,
 - adsorption,
 - condensation, and
 - incineration (combustion).

} Recovery techniques

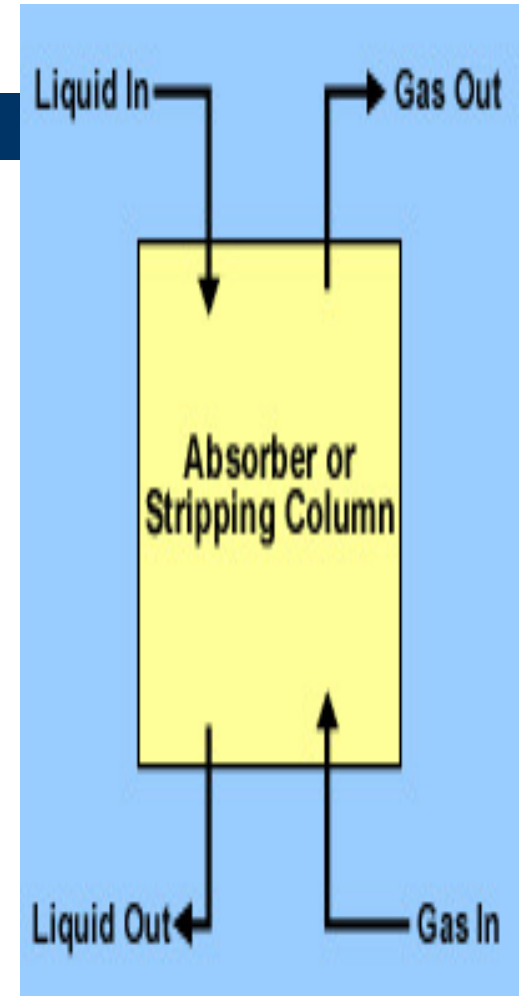
→ Destruction of pollutants

The choice of control technology depends on

- the pollutant(s) that must be removed,
- the removal efficiency required,
- pollutant gas stream characteristics, and
- specific characteristics of the site.

Absorption

- Removal of one or more selected components from a gas mixture by absorption. Water is the most commonly used absorbent liquid.
- Types of gas absorption equipment include spray towers, packed columns (most commonly used) and venturi scrubbers.
- used to recover products or to purify gas streams that have **high concentrations** of organic compounds.
- One potential problem - converts an air pollution problem to a **water pollution problem**.



Points to note **on Absorption**

- Gas solubility in SELECTIVE liquid, which must be:
 - Low vapour pressure
 - Noncorrosive
 - Nontoxic
 - Nonflammable
 - Chemically stable
 - Readily available

WATER

On Absorption – contd.

- Examples of gases that use water as solvent are: HCL and HF
- To enhance removal of certain gases, water is treated with an acid or a base, eg in SO₂ removal
- SO₂ slightly soluble in water but readily soluble in alkaline solution
 - Alkaline solvents (/) are:
 - Regenerable: Na₂SO₃, and NaHPO₄
 - Non-regenerable: CaCO₃

Design considerations - Absorption

- Solvent selection (gas solubility, volatility, cost, viscosity, stability, toxicity, corrosiveness etc.)
- Equilibrium data evaluation
- Operating data estimation
- Column selection (packed, spray, plate etc.)
- Calculation of column diameter
- Column height or number of plates
- Determination of pressure drop thru the column



Preliminary sizing (rough estimate) of an absorber for gas cleanup

Guideline to roughly estimating the size (diameter and height) of a randomly packed absorber to achieve a given degree of gas cleanup.

Assumption

- Absorbent is water or any liquid with similar properties
- Pollutant has a strong affinity with the absorbent

- **Step 1-** **Estimate diameter of column**
Superficial velocity should be 1 – 2 m/s
- **Step 2-** **Choose a packing size for the column**
If $D \sim 1\text{m}$, then use packing with diameter 1in
If $D < 1\text{m}$, then use smaller packings
If $D > 1\text{m}$, then use larger packings

Preliminary sizing (rough estimate) of an absorber for gas cleanup - contd

- Step 3 - Estimate a height for the column

$$\text{Height} = H_{OG} * N_{OG} * \text{safety factor (1.25 - 1.50)}$$

no equilibrium data, and assumption is pollutant is extremely soluble in the absorbent, therefore

$$N_{OG} = \int_{y_2}^{y_1} \frac{dy}{y - y_e} = \int_{y_2}^{y_1} \frac{dy}{y} = \ln y \Big|_{y_2}^{y_1} = \ln \frac{y_1}{y_2}$$

For H_{OG} since solvent is water or something similar

Packing diameter, in	H_{OG} for plastic packing, ft	H_{OG} for ceramic packing, ft
1.0	1.0	2.0
1.5	1.25	2.5
2.0	1.5	3.0
3.0	2.25	4.5
3.5	2.75	5.5

To increase efficiency of absorption

- **Increase pressure** — as pressure of the system increases, more absorbate is driven from the gas phase to the liquid phase
- **Decrease temperature** — as temperature of the system increases, the amount of pollutant absorbed decreases as more absorbate are driven from the liquid to the gas phase.
- **Other factors are**
 - Increase height of packing
 - Reduce gas flow rate
 - etc., etc.

Adsorption

- This occurs when the exhaust stream is brought into contact with a prepared solids and certain gases are selectively retained on the surface or in the pores of the solid.
- Binding forces are usually weak and reversible.
- Considered for low concentration gases from exhaust streams



Adsorption – contd.

Prepared solids – Adsorbents (finely divided solids with large surface area per unit weight)

Adsorbents	Solid preparation	Surface area
Activated carbon	Carbon material (coal, wood, bone, coconut shells etc.) subjected to heat –live steam @ 900°C-treatment under reduced pressure. Used for Odour Control	~ 1000m ² /g
Silica gel	Sodium silicate reacted with sulphuric acid Used for humidity control	~ 600m ² /g
Alumina	Hydrated aluminium oxide Used for drying gases and air	~ 200m ² /g
Molecular sieves	Dehydrated zeolites (alkali –metal silicates aluminosilicates)	**

Adsorption – contd.

- **Carbon adsorption** (uses activated carbon to control and/or recover gaseous pollutant emissions)
 - Activated carbon is the universal standard removal of trace organic contaminants from gaseous streams
 - Removal efficiencies 95% - 99%
 - used where the recovered organics are valuable.
e.g. in the recovery of perchloroethylene, a compound used in the dry cleaning process.

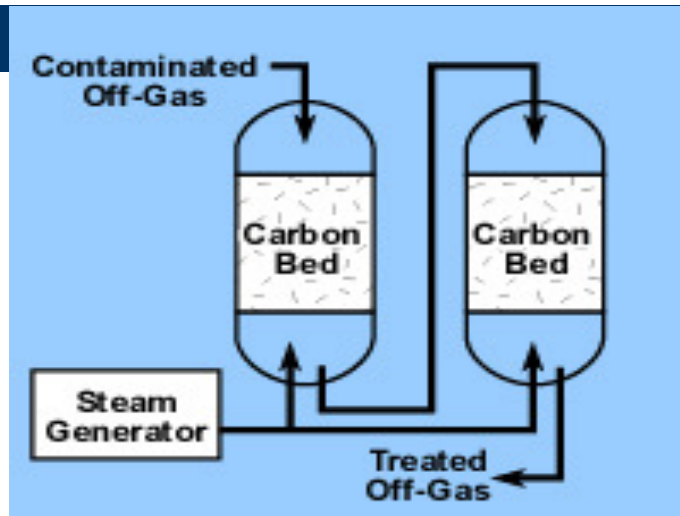
Adsorption – contd.

Carbon adsorption systems are either

regenerative

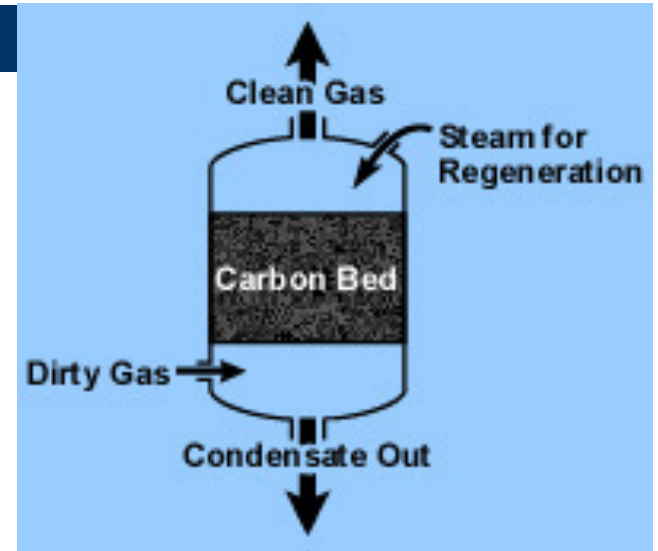
or

non-regenerative



contains more than one carbon bed. As one bed actively removes pollutants, another bed is being regenerated for future use. Steam is used to purge captured pollutants from the bed to a pollutant recovery device.

Used for relatively high conc. of pollutant

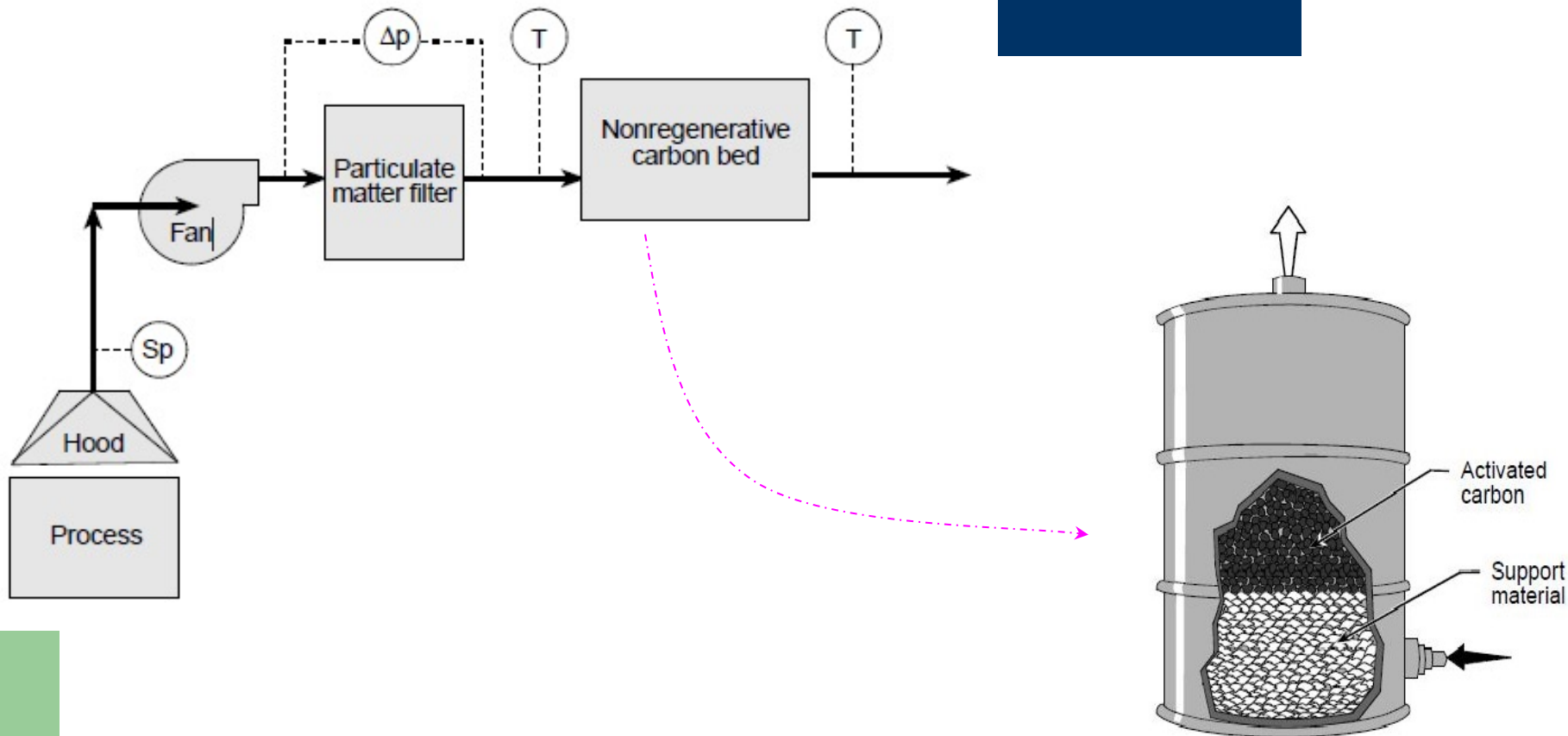


the spent carbon is disposed of when it becomes saturated with the pollutant

Used when the pollutant concentration is extremely low.

Converts the air pollution problem to a **solid waste problem**

Flowchart of a simple nonregenerative adsorber



Adsorption – contd.

Pretreatments before adsorption

- **Water must be removed from gas stream**
adsorbents (except activated carbon) preferentially select water before any other gas
- **Gas stream must be cooled**
adsorbent's activity is regenerated at moderately high temperatures , i.e. desorption takes place - adsorption ability is destroyed at these temperatures:
 - 150°C for activated carbon
 - 600°C for molecular sieves
 - 400°C for silica gel
 - 500°C for activated alumina
- **Particulates must be removed**
particulates can block the pores and spaces between adsorbents

Adsorption – contd.

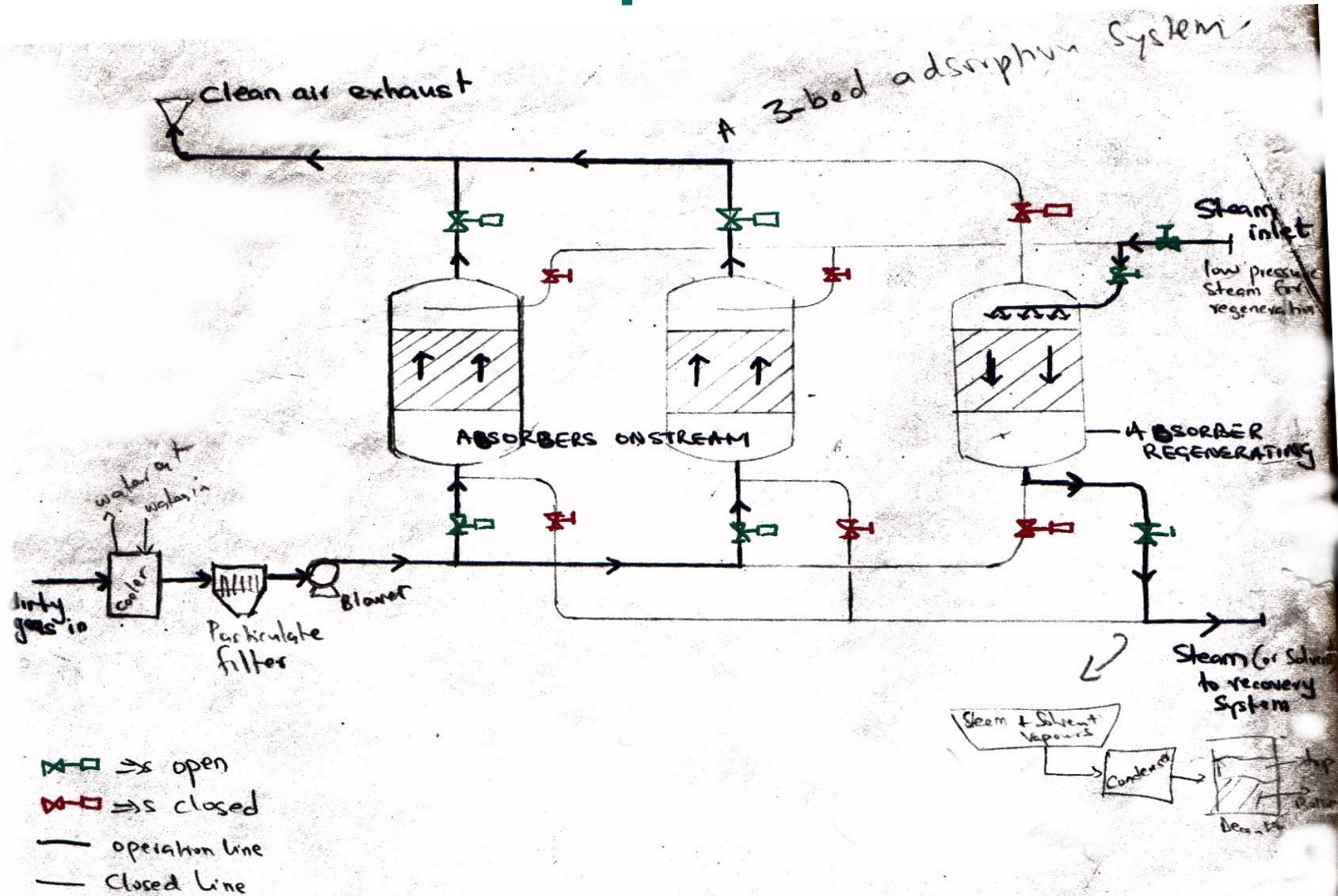
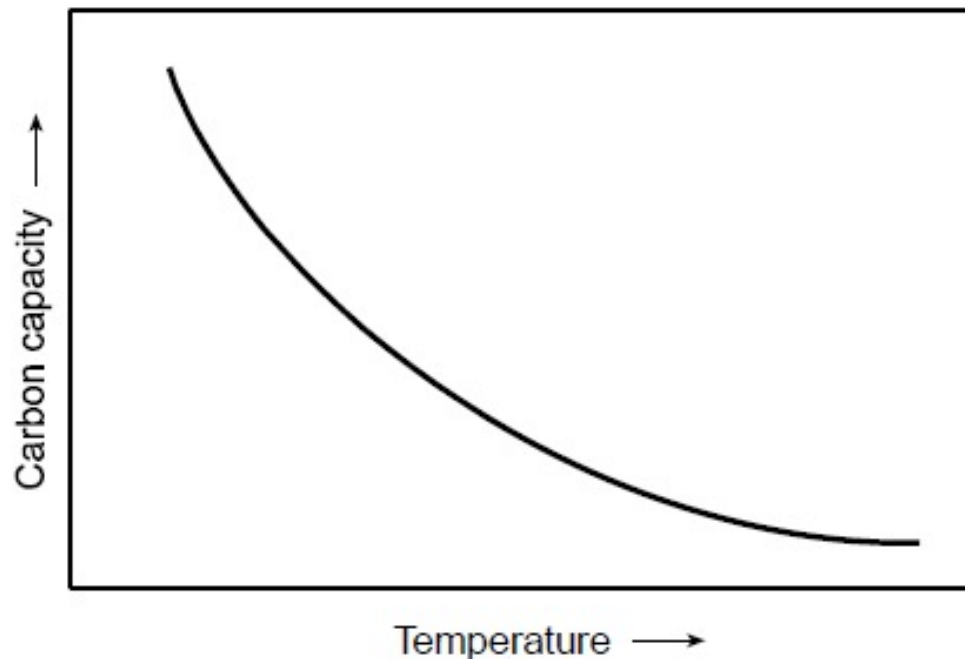


Figure showing that the capacity of an adsorbent decreases as the temperature of the system increases,

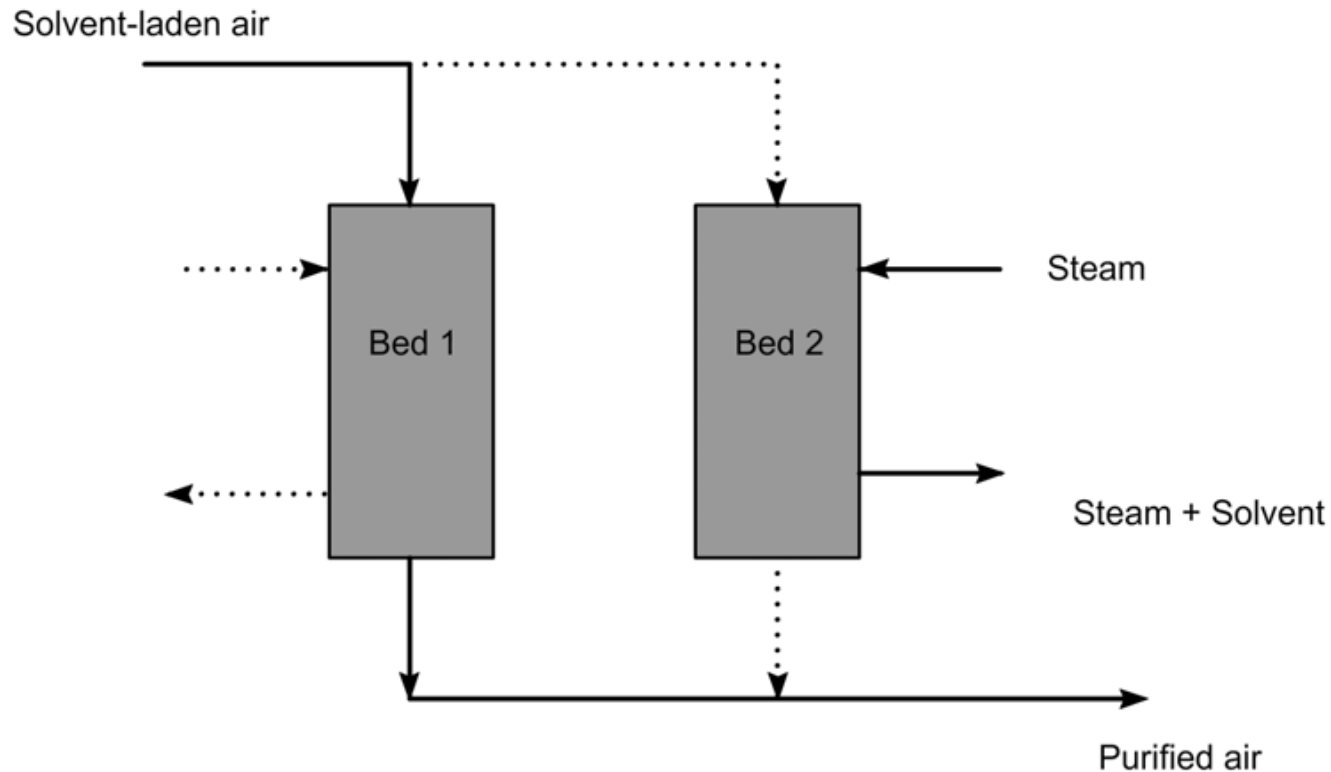


Carbon capacity versus gas stream temperature.

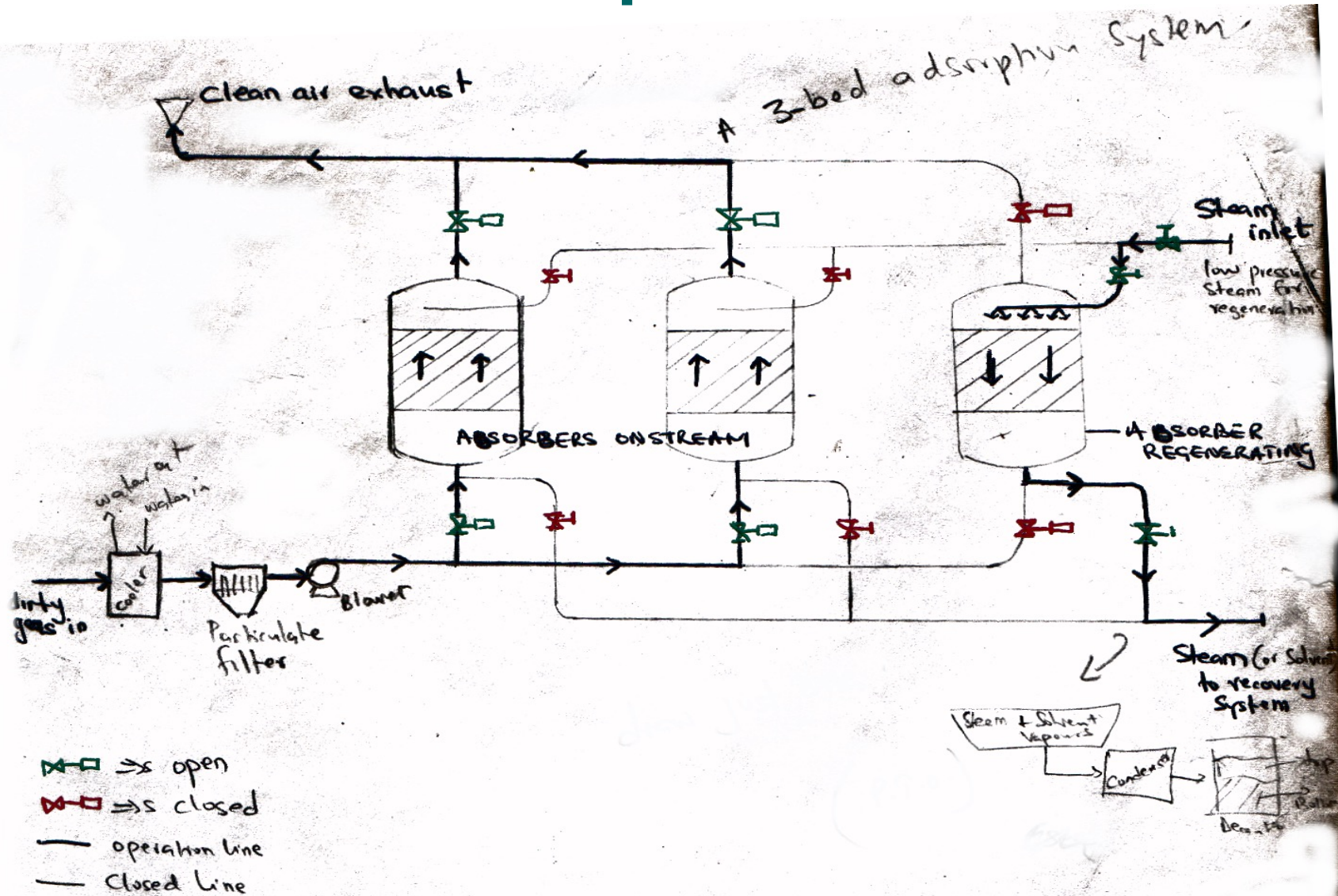
As a general rule, adsorber temperatures are kept below 130°F (54°C) to ensure adequate bed capacities.

Temperatures above this limit can be avoided by cooling the exhaust stream to be treated.

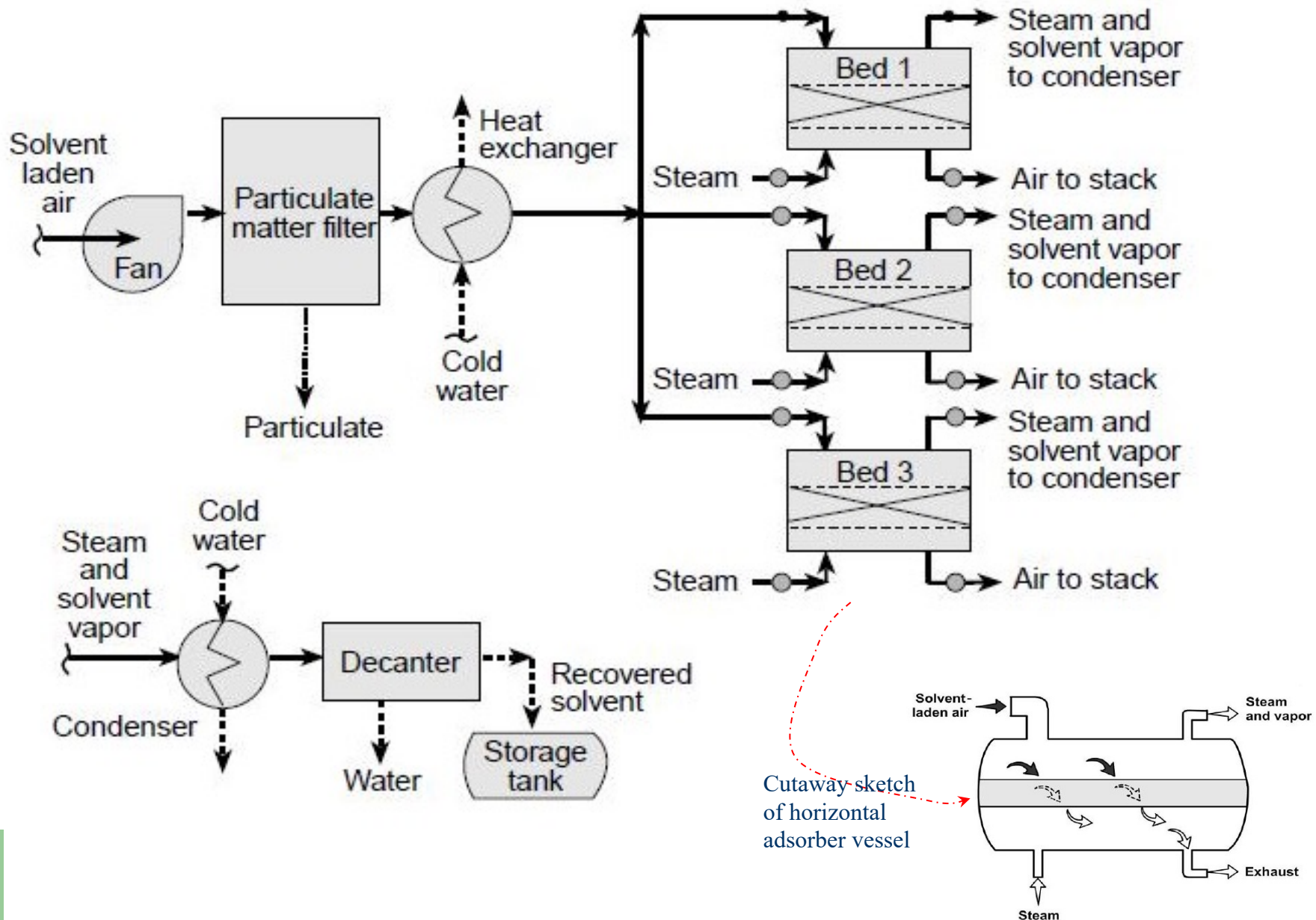
Two-bed adsorption system.



Adsorption – contd.

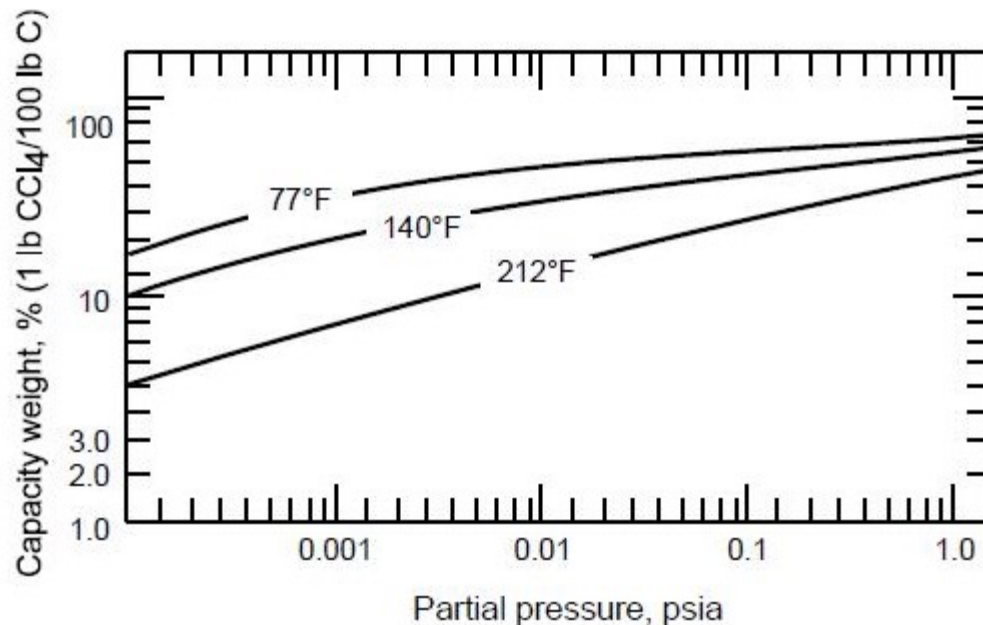


Multi-bed, fixed-bed-type adsorption system



Design considerations - Adsorption

- Adsorbent selection
- Quantity of adsorbent required-
 - Adsorption-Capacity Relationships adsorption isotherm



Adsorption isotherm for carbon tetrachloride on one specific commercial activated carbon adsorbent product

Adsorption capacity may be presented in many different units, with pounds of adsorbate per 100 pounds of adsorbent being typical

Design considerations – Adsorption contd

A dry cleaning process exhausts a 15,000 scfm air stream containing 680 ppm carbon tetrachloride. Given that the exhaust stream is at approximately 140°F and 14.7 psia, determine the saturation capacity of the activated carbon.

STEP 1 - Obtain partial pressure

$$y = 680\text{ppm} = 0.00068$$

$$P^* = yP = (0.00068) \times (14.7\text{psia}) = 0.010\text{psia}$$

STEP 2 - Obtain adsorption capacity from isotherm

This is read off the plot in the previous slide -at a partial pressure of 0.01 psia and a temperature of 140°F, the carbon capacity is read as approximately 45 lb CCl₄/ 100 lb C, or 45%.

This is also equal to 45 gm CCl₄/100 gm C

Still on adsorption capacity

Three important terms are used in actual systems to describe the capacity of the adsorbent bed are:

- **Breakthrough capacity** capacity of the bed at the time where unadsorbed gas begins to be emitted.
- **Saturation capacity** (or equilibrium capacity) - maximum amount of gas that can be adsorbed per unit weight of carbon - the capacity read from the adsorption isotherms.
- **Working capacity** this is a fraction of the saturation capacity, often in the range from 0.1 to 0.5 of the saturation capacity, used for design purposes.

(Smaller working capacities increase the amount of carbon required. The designer selects the appropriate fraction for individual systems by balancing the cost of carbon and adsorber operation versus preventing breakthrough)

Still on adsorption capacity

Recall the case on slide 43 - Estimate the amount of carbon required if the adsorber operates on a four-hour cycle. Use a working capacity of 25% of the saturation capacity. M.Wt. CCl_4 is 154

Step 1. Compute the flow rate of CCl_4 .

$$Q_{\text{CCl}_4} = 15,000 \text{ scfm} \times 0.00068 = 10.2 \text{ scfm } \text{CCl}_4$$

converting to lb/hr

$$\frac{10.2 \text{ ft}^3}{\text{min}} \times \frac{\text{lb-mole}}{385.4 \text{ ft}^3} \times \frac{154 \text{ lb}_m}{\text{lb mole}} \times \frac{60 \text{ min}}{\text{hr}} = 245 \text{ lbm } \text{CCl}_4/\text{hour}$$

For a four-hour cycle: $4 \times 245 = 980 \text{ lbm } \text{CCl}_4$

Step 2. The amount of activated carbon (at saturation) required is:

$$980 \text{ lbm } \text{CCl}_4 \times \frac{100 \text{ lbm carbon}}{45 \text{ lbm } \text{CCl}_4} = 2178 \text{ lbm activated carbon}$$

Based on the working capacity, the actual amount of activated carbon required =
 $4 \times 2178 = \underline{8710 \text{ lbm carbon per four-hour cycle per adsorber}}$

AMOUNT OF ADSORBENT FOR A VOC ADSORBER

Determine the required height of adsorbent in an adsorption column that treats a degreaser-ventilation stream contaminated with trichloroethylene (TCE). Design and operating data are as follows:

Volumetric flow rate of contaminated air: 10,000 std ft³/min
(4.72 m³/s), standard conditions being 60°F and 1 atm

Operating temperature: 70°F (294 K)

Operating pressure: 20 psia (138 kPa)

Adsorbent: activated carbon

Bulk density ρ_B of activated carbon: 36 lb/ft³ (576 kg/m³)

Working capacity of activated carbon: 28 lb TCE per 100 lb carbon

Inlet concentration of TCE: 2000 ppm (by volume)

Molecular weight of TCE: 131.5

The adsorption column is a vertical cylinder with an inside diameter of 6 ft (1.8 m) and a height of 15 ft (4.57 m). It operates on the following cycle: 4 h in the adsorption mode, 2 h for heating and desorbing, 1 h for cooling, 1 h for standby. An identical column treats the contaminated gas while the first one is not in the adsorption mode. The system is required to recover 99.5 percent of the TCE by weight.

Procedure

1. **Determine the actual volumetric flow rate of the contaminated gas stream.** The flow rate as stated corresponds to flow at 60°F and 1 atm, whereas the actual flow q is at 70°F and 20 psia. Accordingly,

$$\begin{aligned} q &= 10,000[(70 + 460)/(60 + 460)][14.7/20] \\ &= 7491 \text{ actual ft}^3/\text{min}, \text{ or } 4.5 \times 10^5 \text{ actual ft}^3/\text{h} (3.54 \text{ actual m}^3/\text{s}) \end{aligned}$$

2. **Calculate the volumetric flow rate of TCE.** This flow rate q_{TCE} equals the inlet concentration y_{TCE} of TCE in the gas times the gas flow rate. Thus,

$$\begin{aligned} q_{\text{TCE}} &= (y_{\text{TCE}})(q) = (2000 \times 10^{-6})(4.5 \times 10^5) \\ &= 900 \text{ actual ft}^3/\text{h} (0.007 \text{ m}^3/\text{s}) \end{aligned}$$

3. **Convert the volumetric flow rate of TCE into mass flow rate.** For the conversion, rearrange the ideal-gas law, bearing in mind that mass equals the number of moles times the molecular weight. Thus $\dot{m}$, the mass flow rate in pounds per hour, equals $q_{\text{TCE}}(PM/RT)$, where P is the pressure, M the molecular weight, T the absolute temperature, and R the gas constant. Accordingly,

$$\dot{m} = (900)(131.5)/(10.73)(70 + 460) = 416.2 \text{ lb/h} (0.052 \text{ kg/s})$$

4. **Determine the mass of TCE to be adsorbed during the 4-h period.** This equals, simply, the required degree of adsorption times the amount of TCE that will pass through the system in 4 h:

$$\text{TCE adsorbed} = (416.2 \text{ lb/h})(4 \text{ h})(0.995) = 1656.6 \text{ lb}$$

5. **Calculate the volume of activated carbon required.** To obtain this volume v_{AC} , divide the amount of TCE to be adsorbed by the adsorption capacity of the carbon, and convert from mass to volume by taking into account the bulk density of the carbon:

$$\begin{aligned} v_{\text{AC}} &= (\text{TCE to be adsorbed})/(28 \text{ lb TCE}/100 \text{ lb carbon})(\text{bulk density}) \\ &= (1656.6)/(28/100)(36) = 164 \text{ ft}^3 (4.64 \text{ m}^3) \end{aligned}$$

6. **Find the required height of the carbon in the adsorber.** This height z equals the volume of carbon divided by the cross-sectional area of the column:

$$z = 164/[\pi(D^2/4)] = 164/[\pi(6^2/4)] = 5.8 \text{ ft} (1.77 \text{ m})$$

Factors Affecting Adsorption System Performance

- **Temperature** - adsorbent capacity decreases as temp increases. As a general rule, adsorber temperatures are kept below 130°F (54°C)
- **Pressure** - adsorption capacity increases with an increase in the partial pressure of the gas, which is proportional to the total pressure of the system. Any increase in pressure will increase the adsorption capacity.
- **Gas Velocity** - gas velocity through the adsorber determines the residence time. this directly affects capture efficiency. The lower the gas velocity (the longer the contact time) through the adsorbent bed, the greater is the probability of a contaminant molecule reaching an available site. For a removal efficiency of 90% adsorbers are designed for a maximum gas velocity of 100 ft/min (30 m/min).
- **Humidity**- at high relative humidity (> 50%) the water molecules increases to the extent that they begin to compete with the hydrocarbon molecules for active adsorption sites, thus reducing the capacity and efficiency of the adsorption system